Doing Language Science:

Phonetic Analysis of Vowels

PLATFORM | SP2026

The basics



Today's plan:

- Quick recap
- bit of new info on Acoustics
- Making vowel plots

Phonetics

- **Phonetics:** the study of the sounds of human language
- Or, more realistically, the study of the physical properties of language
- Linguists that specialize in Phonetics are called Phoneticians

THE INTERNATIONAL PHONETIC ALPHABET (revised to 2005)

CONSONANTS (PULMONIC)

© 2005 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Where symbols appear in pairs, the one to the right represents a voiced consonant. Shaded areas denote articulations judged impossible.

CONSONANTS (NON-PULMONIC)

Clicks		Voiced implosives		Ejectives	
ʘ	Bilabial	ɓ	Bilabial	ʼ	Examples:
ǀ	Dental	ɗ	Dental/alveolar	pʼ	Bilabial
ǃ	(Post)alveolar	ʄ	Palatal	tʼ	Dental/alveolar
ǂ	Palatoalveolar	ɠ	Velar	kʼ	Velar
ǁ	Alveolar lateral	ʛ	Uvular	sʼ	Alveolar fricative

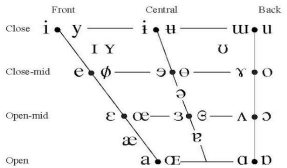
OTHER SYMBOLS

ʍ	Voiceless labial-velar fricative	ɕ ʑ	Alveolo-palatal fricatives
w	Voiced labial-velar approximant	ɺ	Voiced alveolar lateral flap
ɥ	Voiced labial-palatal approximant	ʃʒ	Simultaneous ʃ and x
ʜ	Voiceless epiglottal fricative		
ʢ	Voiced epiglottal fricative		
ʡ	Epiglottal plosive		

Affricates and double articulations can be represented by two symbols joined by a tie bar if necessary.

k͡p t͡s

VOWELS



Where symbols appear in pairs, the one to the right represents a rounded vowel.

SUPRASEGMENTALS

ˈ	Primary stress
ˌ	Secondary stress ˌfoʊnəˈtɪʃən
ː	Long eː
ˑ	Half-long eˑ
˘	Extra-short ĕ
|	Minor (foot) group
‖	Major (intonation) group
.	Syllable break ɹi.ækt
‿	Linking (absence of a break)

DIACRITICS

Diacritics may be placed above a symbol with a descender, e.g. n̥

◌̥	Voiceless	n̥ d̥	◌̤	Breathy voiced	b̤ a̤
◌̬	Voiced	s̬ t̬	◌̰	Creaky voiced	b̰ a̰
◌ʰ	Aspirated	tʰ dʰ	◌̼	Linguolabial	t̼ d̼
◌̹	More rounded	ɔ̹	◌ʷ	Labialized	tʷ dʷ
◌̜	Less rounded	ɔ̜	◌ʲ	Palatalized	tʲ dʲ
◌̟	Advanced	u̟	◌ˠ	Velarized	tˠ dˠ
◌̠	Retracted	e̠	◌ˤ	Pharyngealized	tˤ dˤ
◌̈	Centralized	ë	◌̴	Velarized or pharyngealized	ɫ
◌̽	Mid-centralized	e̽	◌̝	Raised	e̝ (ɹ̝ = voiced alveolar fricative)
◌̩	Syllabic	n̩	◌̞	Lowered	e̞ (β̞ = voiced bilabial approximant)
◌̯	Non-syllabic	e̯	◌̘	Advanced Tongue Root	e̘
◌˞	Rhoticity	ɚ a˞	◌̙	Retracted Tongue Root	e̙
◌̪	Dental	t̪ d̪	◌̺	Apical	t̺ d̺
◌̻	Laminal	t̻ d̻	◌̃	Nasalized	ẽ
			◌ⁿ	Nasal release	dⁿ
			◌ˡ	Lateral release	dˡ
			◌̚	No audible release	d̚

TONES AND WORD ACCENTS

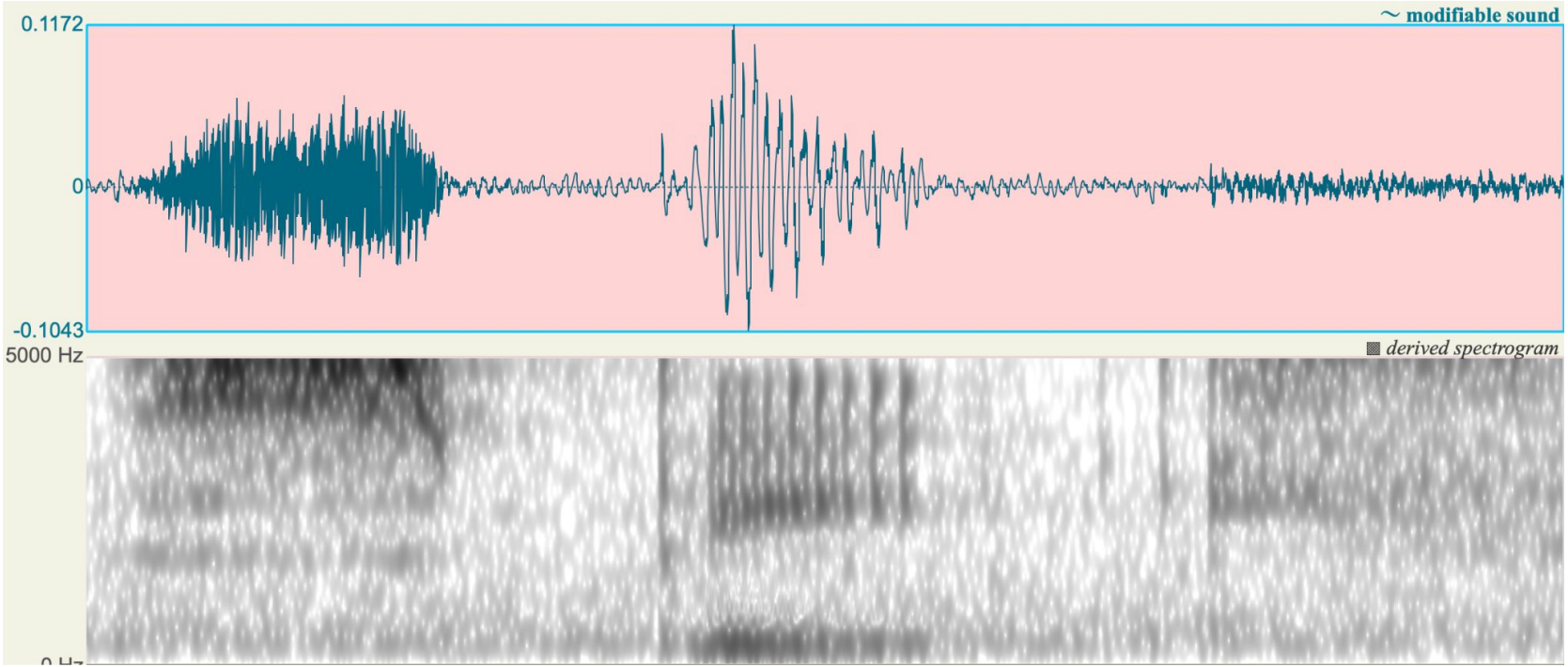
LEVEL		CONTOUR	
é or ˥	Extra high	ě or ˩˥	Rising
é ˦	High	ê ˥˩	Falling
ē ˧	Mid	ě ˧˥	High rising
è ˨	Low	e̋ ˨˩	Low rising
ȅ ˩	Extra low	ê ˩˥	Rising-falling
↓	Downstep	↗	Global rise
↑	Upstep	↘	Global fall

IPA

- In other words, the IPA is a tool that helps us separate speech from spelling in order to talk about sounds empirically– *as they really are*
    - (or like, much closer to how they really are)

cats= k æ t s
bees=b i z

Acoustic Information

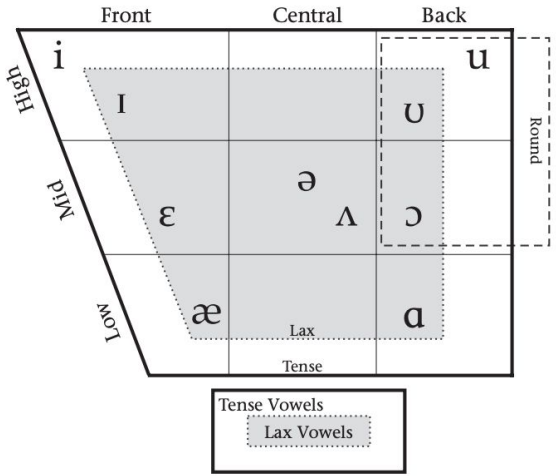


Cs and Vs

CONSONANTS are sounds in human speech that CONSTRICT AIRFLOW somewhere in the vocal tract.

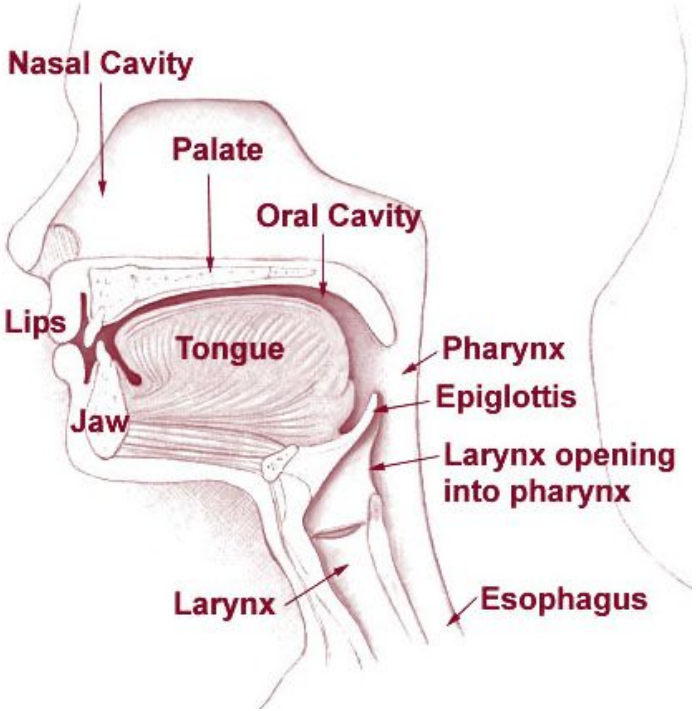
VOWELS are sounds that have a relatively OPEN VOCAL TRACT with no built-up air pressure anywhere

The Vowel Space



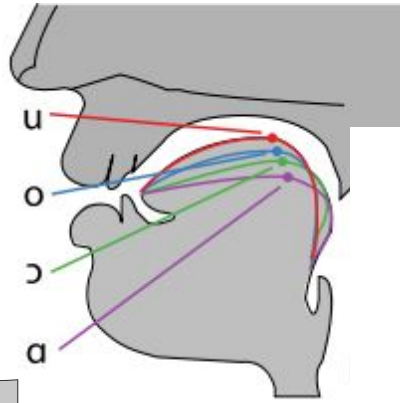
- On the IPA chart, we put vowels on a quadrilateral, and talk about them in terms of height, frontness, etc.

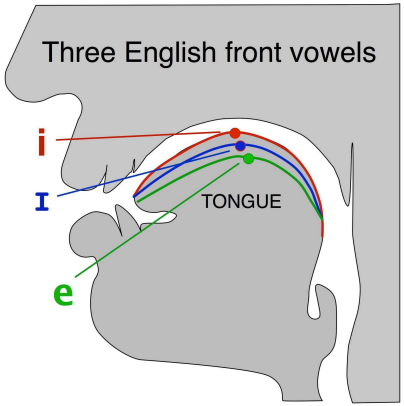
The Vowel Space

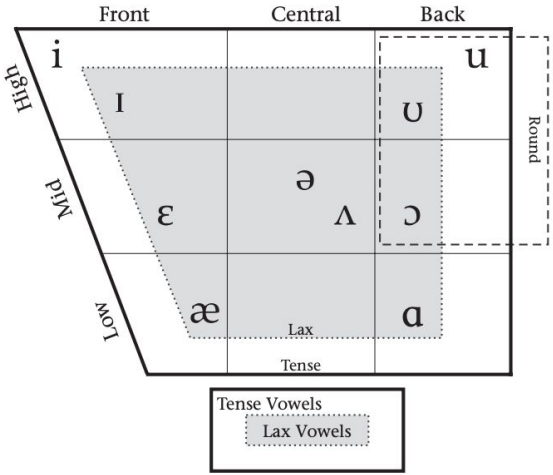


- On the IPA chart, we put vowels on a quadrilateral, and talk about them in terms of height, frontness, etc.
- This shape mostly corresponds to the oral cavity our tongues can move around in!
- So articulation matches up with our perceptual labels, sweet

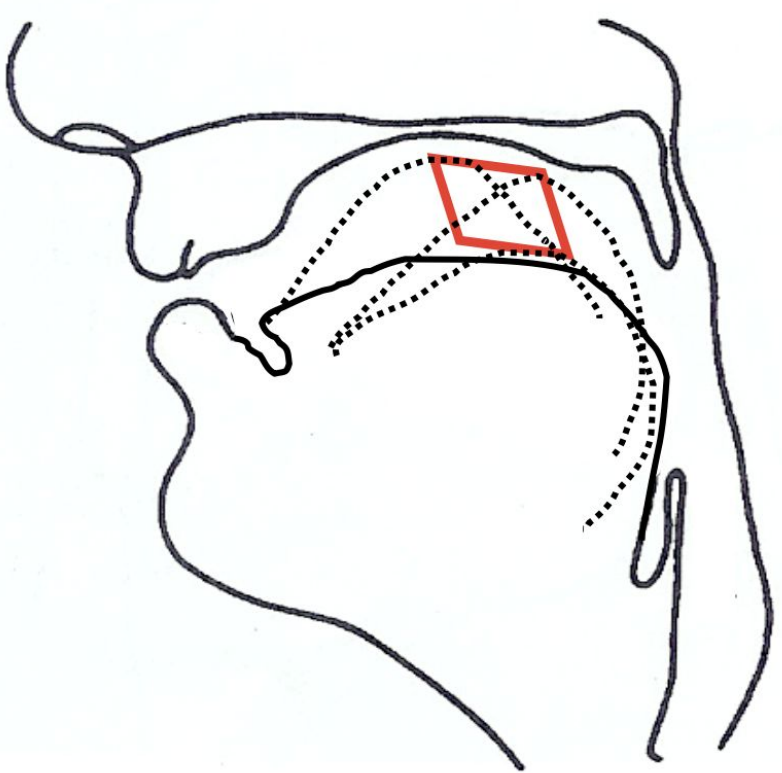
Tongue height & Tongue position

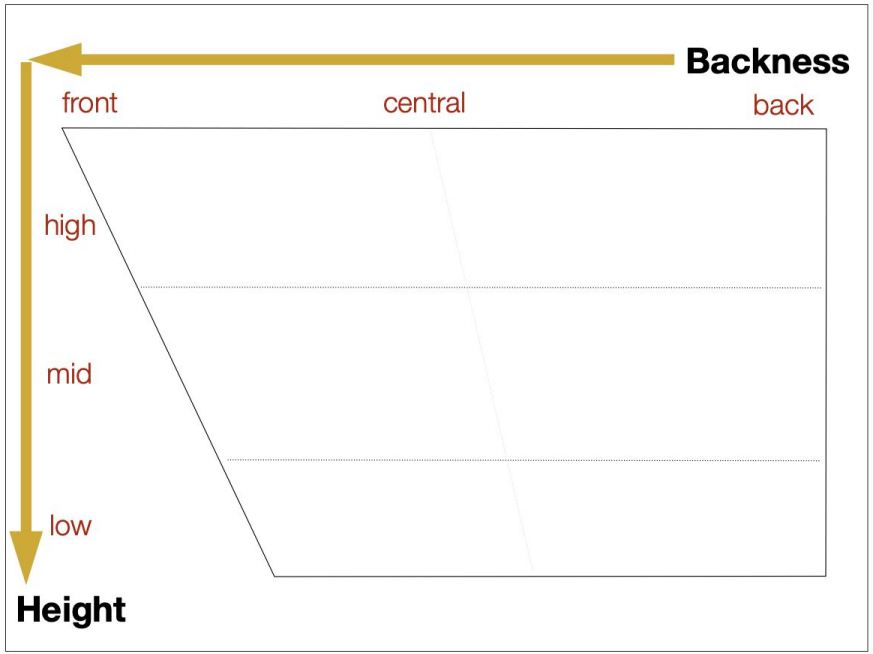






The Vowel Space



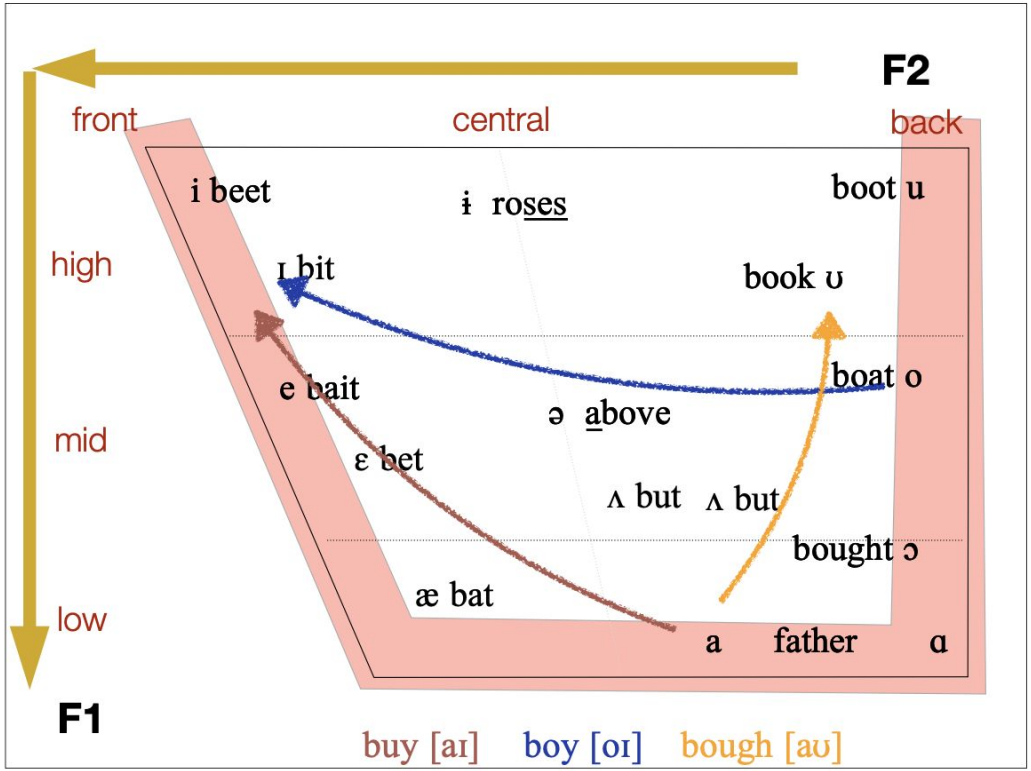


English Spelling is (also):





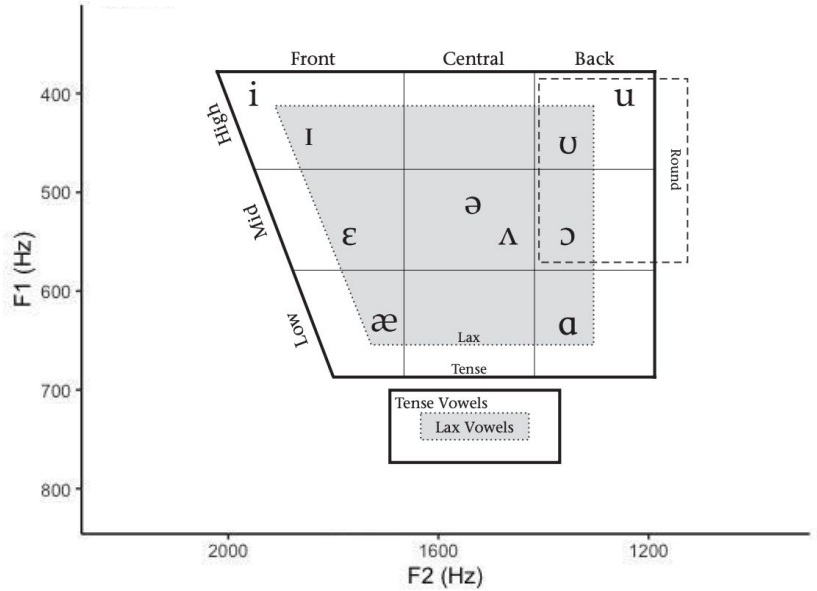
The truth about English Vowels:



Acoustics

- The vowel space **also** lines up with acoustic measurements that we'll call the first and second formant frequencies (F1 and F2)

Isaac Newton actually recognized the relation between vowel qualities and resonances, but we kind of only really did anything with that info around 1940s



F1 and F2

- So all we need to position vowels in space are these two values!

- But what are they? And how do we find them?
- Formants =frequency peaks amplified by the meat tube we use to talk. The frequencies that get amplified depend on how we shape the tube
- This is why [a] is different from [i]; also why a trombone and a trumpet sound different

Formants, a different explanation:

- So vowels differ in their formants (F1 and F2 mostly for vowels), and so do instruments, and voices, tone quality, timbre, etc
- In other words, when we decompose a sound into its formants, it's kind of like unmixing a color of paint.
- Trying to determine what colors were blended together into the single color that we *actually* see



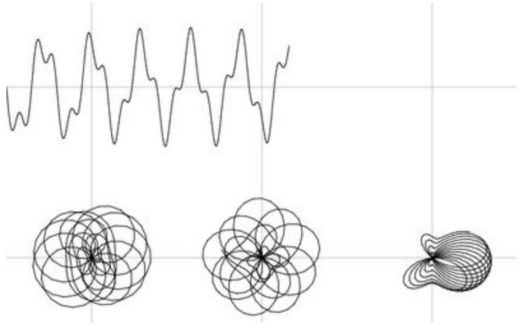
F1 and F2

- So all we need to position vowels in a space are these two formant values

- We can calculate formants using software like praat which shows us **spectrograms** of sound

- When we have formant measures for a vowel, we can plot those as ordered pairs, <(X,Y)=(F2, F1)>, and we end up with a vowel space

Side note:

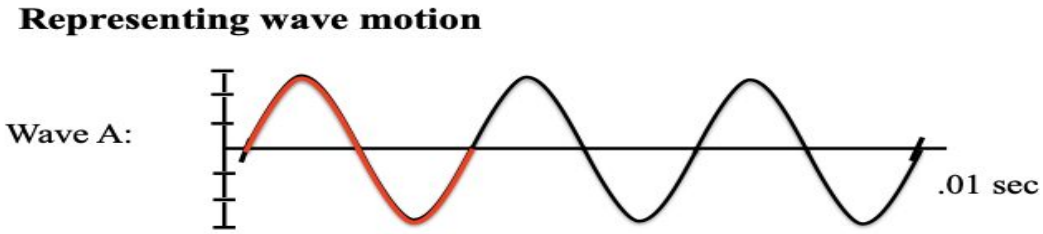


- We *could* do it by hand in a couple different ways if we didn't have software to do it for us
- You COULD use a crazy big speaker with a pencil taped to the back of it attached to the top of a rotating flag pole, and then put graph paper behind it to get a frequency spectrum, or you could just calculate a fourier transform of the wave form into a frequency spectrum
- and then you could **use** those frequency spectra to estimate resonances of our vocal tract and make a *spectrogram*
- But we won't! and we won't even have to do the math by hand!

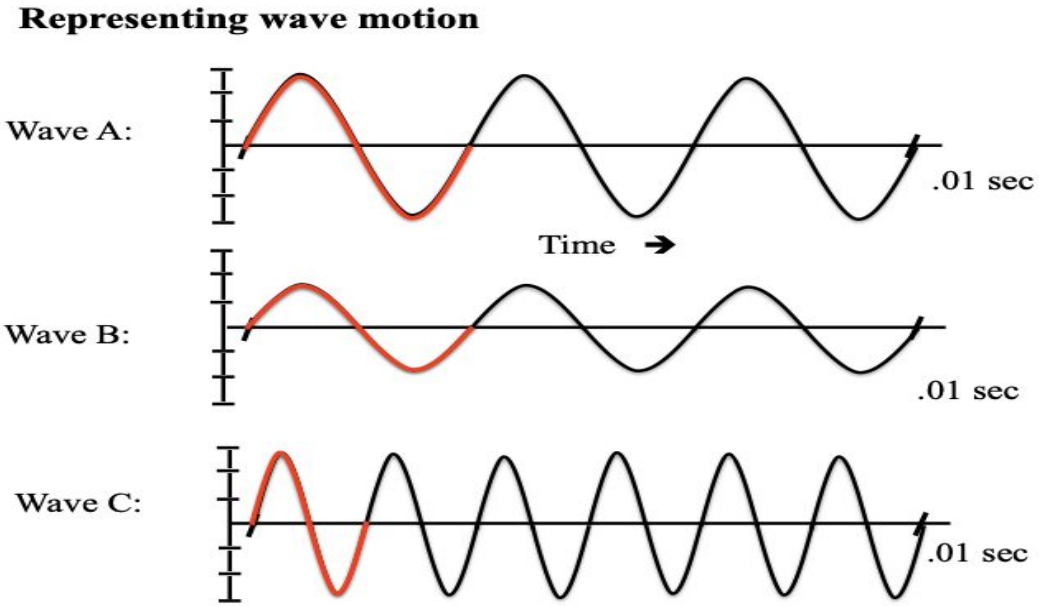
$$X_k = \frac{1}{N}\sum_{n=0}^{N-1} x_n e^{i2\pi k \frac{n}{N}}$$

To find the energy at a particular frequency, spin your signal around a circle at that frequency, and average a bunch of points along that path.

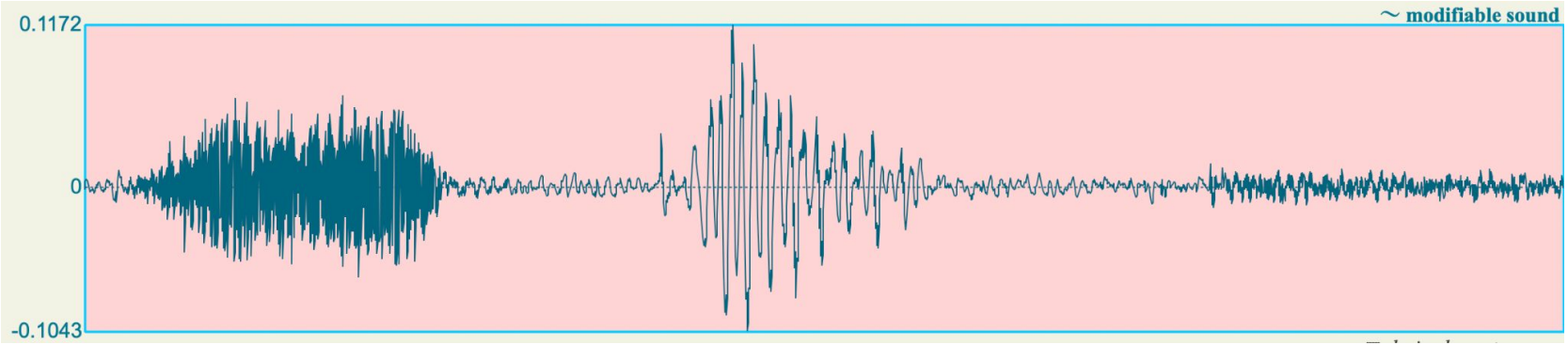
Basic Acoustics: Vibration & Waves



Basic Acoustics: Vibration & Waves

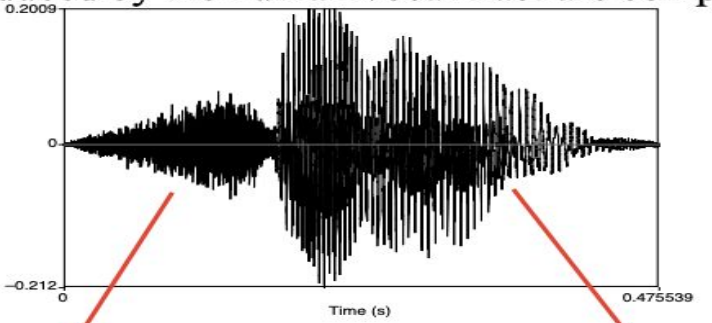


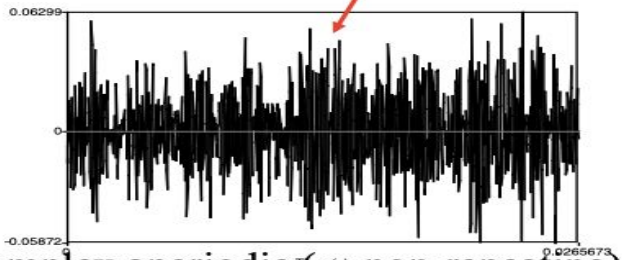
But speech looks like this:



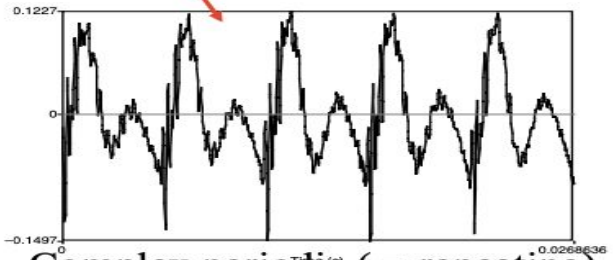
COMPLEX WAVES

Sound waves produced by the human vocal tract are complex:



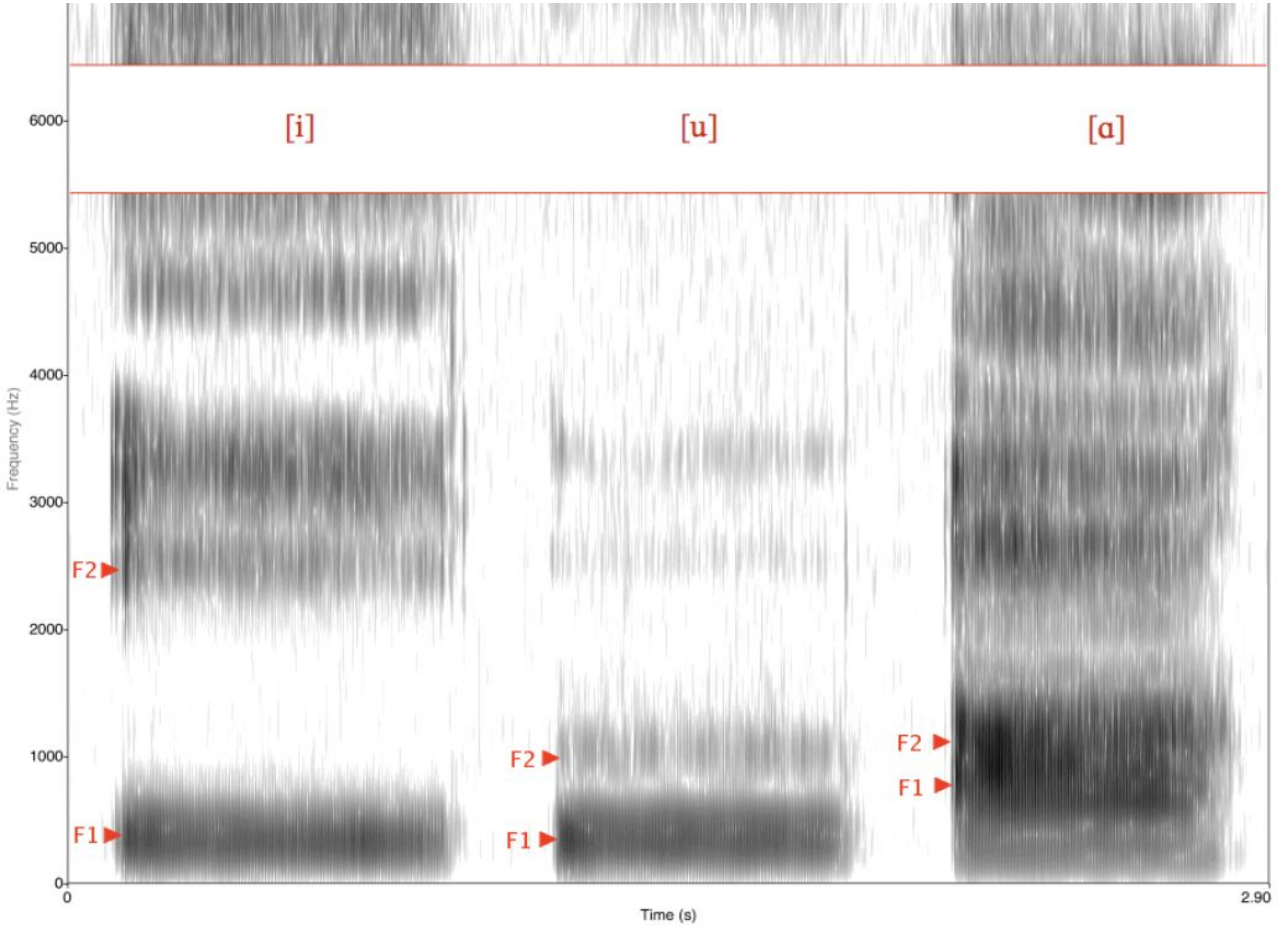


Complex aperiodic (= non-repeating)



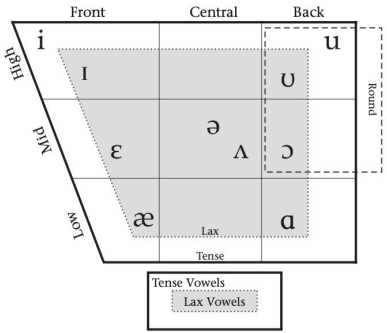
Complex periodic (= repeating)

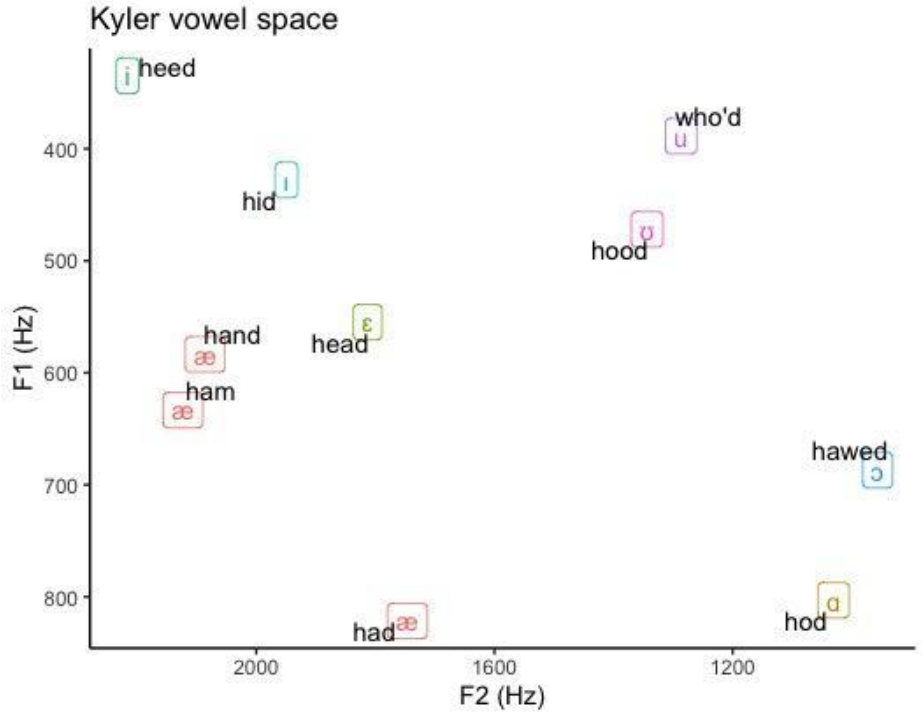
F1 and F2


[i]
[u]
[ɑ]
6000
5000
4000
3000
2000
1000
0
Frequency (Hz)
F2
F1
F2
F1
F2
F1
0
2.90
Time (s)

Vowel Plots

- What do you notice when comparing these two vowel spaces?





Doing a phonetic analysis of your vowels

Stuff you need to make a vowel plot:

- Functioning human vocal tract or equivalent
- Computer with microphone to record
- Praat

Doing a phonetic analysis of your vowels

What you do:

- Record words
- View recording in Praat
- Zoom in to the vowel
- Click in the middle of the repeating shapes where they’re stable
- Click ‘formants’ > ‘formant listing’ and record F1 and F2, paste into the script
- Repeat

Doing a phonetic analysis of your vowels

Specifics in Praat:

- ‘New’>’record mono sound’>’record’>’save to list & close’
- ‘View and edit’
- Select vowel then click ‘sel’ and repeat until its zoomed in
- Click in the middle of the repeating shapes where they’re stable
- Click ‘formants’ > ‘formant listing’, get numbers for F1 and F2, paste into the script (note: f2 is the FIRST number in the script)
- Repeat

I’ll walk you through this now so you can see

Here are the words you'll say (in this order!):

- heed
- hid
- hayed
- head
- had
- hod
- hoed
- hood
- who'd?

Doing a phonetic analysis of your vowels

There's a script I wrote called 'plotvowels.praat' on the desktops.

All you need to do with it is open it, and paste in **F2 and F1** in the bottom part for each vowel, then click run

You'll get a plot!

F2 F1

```
Text... 2790  Centre 310 Half i
Text... 2480  Centre 430 Half \ic
Text... 2620  Centre 540 Half e
Text... 2330  Centre 610 Half \ef
Text... 2050  Centre 860 Half \ae
Text... 1520  Centre 950 Half a
Text... 920   Centre 590 Half o
Text... 1160  Centre 470 Half \hs
Text... 950   Centre 370 Half u
```

heed	i	i
hid	ɪ	\ic
hayed	e	e
head	ɛ	\ef
had	æ	\ae
hod	ɑ	a
hoed	o	o
hood	ʊ	\hs
who’d?	u	u

Variation

Just like these plots->

the vowels you plot are going to be specific to *you.*

They'll be *similar* to other peoples' but not the same

